$$E(aX+b) = a\,E(X) \pm b$$

$$E(aX \pm b) = \int_{-\infty}^{\infty} (ax \pm b) f(x)\,dx$$

$$= a \int_{-\infty}^{\infty} x f(x)\,dx \pm b \int_{-\infty}^{\infty} f(x)\,dx$$

$$= a\,E(X) \pm b \times 1 \qquad \text{as} \int_{-\infty}^{\infty} f(x)\,dx = 1$$

$$= a\,E((X) \pm b)$$

Remarks:

1. **Statement:** $E(aX+b) = aE(X)+b$ where a and b are constants.

Proof: $E(aX+b) = E(aX)+E(b)$ by property 3

$= aE(X)+b$ by property 2

Similarly $E(aX-b) = aE(X)-b$

2. **Statement:** $E\left(\frac{ax+b}{c}\right) = \frac{[aE(x)+b]}{c}$

Proof: $E\frac{(aX+b)}{c} = \frac{E(aX+b)}{c}$

$= \frac{[aE(x)+b]}{c}$

3. **Statement:** $E(X - \overline{X}) = 0$

Proof: $E(X - \overline{X}) = E(X) - E(\overline{X})$ since $E(X) = \overline{X}$

$= \overline{X} - \overline{X} = 0$ $\overline{X}$ is a constant

Example 9.15

Find the expectation of the sum of the number obtained on throwing two dice.

Solution:

Let X&Y denote the number obtained on the I and II die respectively. Then each of them is a random variable which takes the value 1,2,3,4,5 and 6 with equal probability $\frac{1}{6}$